

Manankumar Thakkar

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SUMMARY

Software Engineer experienced in designing scalable, distributed systems in Java and Spring Boot and building generative AI agents and LLM tools. Skilled in design patterns, reliability and scaling, multi-threaded performance, Kubernetes, and the full development life cycle. Proven in production: cut API latency **60%** and prevented a **6x data collision** during multi-tenant expansion.

TECHNICAL SKILLS

Languages: Java, Python, SQL, TypeScript, JavaScript
Backend: Spring Boot, Microservices, Distributed systems, REST APIs, Concurrency, Kafka, Redis, Flask
AI / LLM: Generative AI agents, LLM integration, MCP / tool use, LangChain, Retrieval-grounded QA, Prompt engineering
Cloud & Data: Kubernetes, Docker, GCP (BigQuery, GCS), Azure SQL, AWS, Terraform, Liquibase
SDLC & Ops: CI/CD, Git, Code reviews, JUnit, Grafana, Prometheus, Splunk, SonarQube, Snyk

EXPERIENCE

Walmart Global Tech Bentonville, AR
Software Engineer III 02/2025 – Present

- Architected the Vision Center feature in Java/Spring Boot so the Inventory Reservation Service could set loop policy per location, and led the migration from percentage- to seconds-based loop metrics, expanding advertiser booking flexibility **40%**.
- Diagnosed a production N+1 query bottleneck across distributed microservices and fixed it with concurrent, multi-threaded caching, cutting API latency **60%** and eliminating **10,000+ database calls** per day.
- Caught a key-grain design flaw while integrating multi-tenant international expansion across 7+ partner orgs, and drove a durable country-scoped key over a regional patch, preventing a **6x cardinality collision**.
- Built a generative AI operations assistant (Python, Flask, LangChain) on Walmart's internal LLM Gateway, giving engineers chat access to operational context via a REST API with session-scoped memory and response caching.
- Mentored 3+ engineers on deployments and complex system flows through code reviews, and served as Release Captain for 4+ production deployments across 5+ teams with zero customer-impacting incidents.

Software Engineer 07/2024 – 01/2025

- Extended the store data model and built Java/Spring Boot services for Store Department Hours and Sam's Club Plus operating hours, adding Deli, Bakery, and Service Deli schedules across **4,000+ stores**.
- Enabled targeted in-store ad placement for new retail formats (Grab-N-Go, Sam's Club Cafe) by building location-to-screen-group mapping and filtering across the stores, inventories, and device-commands REST APIs.

George Mason University Fairfax, VA
AI Research Assistant 07/2023 – 07/2024

- Built research tooling powered by LLMs (ChatGPT) that automated and streamlined manual research workflows.
- Provisioned and managed research infrastructure as code with Terraform for reproducible research environments.

Nokia Remote, US
Test Automation Engineer (Co-op) 09/2022 – 06/2023

- Automated testing with Python suites for high-throughput systems, reducing manual testing effort **50%**.
- Integrated tests into Jenkins CI/CD pipelines, cutting defect density **20%** and improving code quality metrics **35%**.

Let It Wag Mumbai, India
Full Stack Developer 06/2019 – 05/2020

- Engineered Node.js services and REST APIs handling real-time data for **13K+ users**, improving backend efficiency **20%**.
- Developed a responsive React UI supporting 100+ user journeys and delivered 90% of projects on time.

AI PROJECTS

AuthForge | Python, MCP, Spring Boot, Istio

- Designed an MCP server that remediates missing-authentication findings in Walmart Java services via config alone.
- Implemented 9 tools spanning diagnosis, endpoint scanning, consumer discovery, config generation, and validation.

Wayfinder (Walmart Global Tech Hackathon) | Python, LLM orchestration, agent routing

- Co-built an agentic system that triages Slack, email, and calendar by intent and routes work to sub-agents.
- Owned the routing layer, query sanitization, and extensibility framework, with one-click human approval gates.

EDUCATION

George Mason University, Fairfax, VA – *Master of Science in Computer Science* 08/2021 – 05/2023
Coursework: Data Structures and Algorithms, Software Modeling and Architectural Design, Data Mining, Advanced NLP